11:54

colab.research.google.com

Weather_Dataset_Analysis.ipynb

File Edit View Insert Runtime Tools Help

Share

ommands + Code + Text ▷ Run all

RAM Disk

```python
import pandas as pd
import matplotlib.pyplot as plt

# 1. Load the weather dataset
url = "https://raw.githubusercontent.com/vega/vega-datasets/main/data/seattle-weather.csv"

df = pd.read_csv(url)

# 2. Display the first five rows
print("WEATHER DATASET - FIRST FIVE ROWS")
print(df.head())

# 3. Display dataset information
print("\nDATASET INFORMATION")
print("Total rows:", df.shape[0])
print("Total columns:", df.shape[1])
print("Column names:", df.columns.tolist())

# 4. Calculate temperature statistics
print("\nTEMPERATURE STATISTICS")
print(df[["temp_max", "temp_min"]].describe())

# 5. Count weather types
print("\nWEATHER TYPE COUNTS")
print(df["weather"].value_counts())

# 6. Plot average temperatures
average_temp = df[["temp_max", "temp_min"]].mean()

average_temp.plot(kind="bar")
plt.title("Average Maximum and Minimum Temperature")
plt.xlabel("Temperature Type")
plt.ylabel("Temperature (Celsius)")
plt.tight_layout()
plt.show()

# 7. Plot weather types
df["weather"].value_counts().plot(kind="bar")
plt.title("Weather Type Distribution")
plt.xlabel("Weather Type")
plt.ylabel("Number of Days")
plt.tight_layout()
plt.show()

# 8. Save dataset
df.to_csv("weather.csv", index=False)

print("\nPROJECT COMPLETED SUCCESSFULLY!")
```

Show hidden output

11:55

colab.research.google.com

```
df["weather"].value_counts().plot(kind="bar")
plt.title("Weather Type Distribution")
plt.xlabel("Weather Type")
plt.ylabel("Number of Days")
plt.tight_layout()
plt.show()

# 8. Save dataset
df.to_csv("weather.csv", index=False)

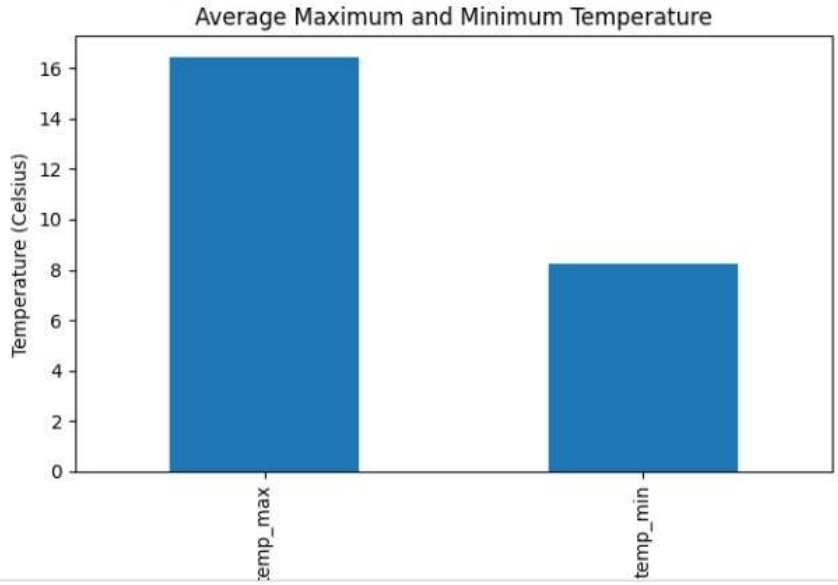
print("\nPROJECT COMPLETED SUCCESSFULLY!")
```

```
WEATHER DATASET - FIRST FIVE ROWS
         date  precipitation  temp_max  temp_min  wind  weather
0  2012-01-01            0.0      12.8       5.0   4.7  drizzle
1  2012-01-02           10.9      10.6       2.8   4.5     rain
2  2012-01-03            0.8      11.7       7.2   2.3     rain
3  2012-01-04           20.3      12.2       5.6   4.7     rain
4  2012-01-05            1.3       8.9       2.8   6.1     rain

DATASET INFORMATION
Total rows: 1461
Total columns: 6
Column names: ['date', 'precipitation', 'temp_max', 'temp_min', 'wind', 'weather']

TEMPERATURE STATISTICS
          temp_max     temp_min
count  1461.000000  1461.000000
mean     16.439083     8.234771
std       7.349758     5.023004
min      -1.600000    -7.100000
25%      10.600000     4.400000
50%      15.600000     8.300000
75%      22.200000    12.200000
max      35.600000    18.300000

WEATHER TYPE COUNTS
weather
rain       641
sun        640
fog        101
drizzle     53
snow        26
Name: count, dtype: int64
```



11:55

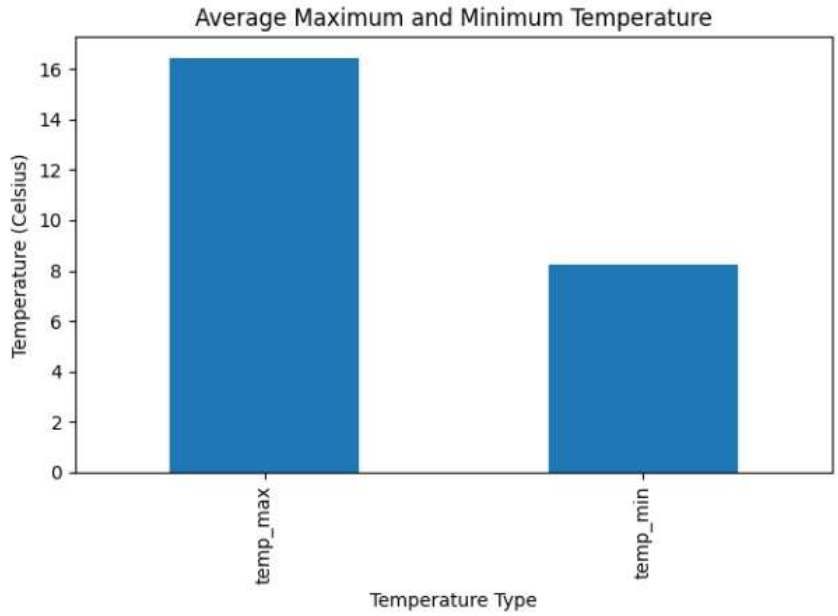
colab.research.google.com

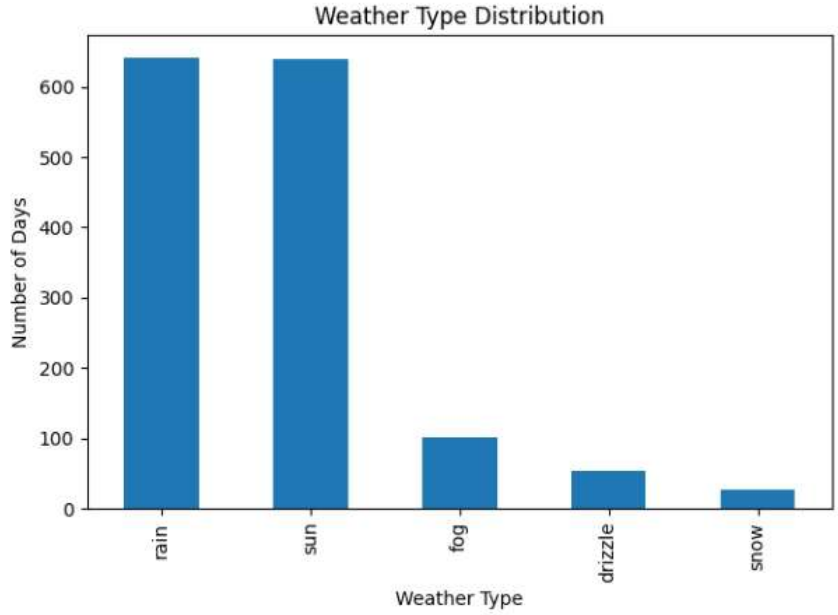
```
df["weather"].value_counts().plot(kind="bar")
plt.title("Weather Type Distribution")
plt.xlabel("Weather Type")
plt.ylabel("Number of Days")
plt.tight_layout()
plt.show()

# 8. Save dataset
df.to_csv("weather.csv", index=False)

print("\nPROJECT COMPLETED SUCCESSFULLY!")
```

```
Name: count, dtype: int64
```






```
PROJECT COMPLETED SUCCESSFULLY!
```